

Daily Review

Booklet p.5-7 — Mass and Weight

- 1 What is the difference between mass and weight?
- 2 A 20 kg crate is on Mars. Calculate its weight.
- 3 What happens to your mass on the Moon?
- 4 What happens to g further from a planet?
- 5 What is the unit of weight?

Mass is matter (kg);
weight is a force (N)

74 N

It stays the same

It decreases

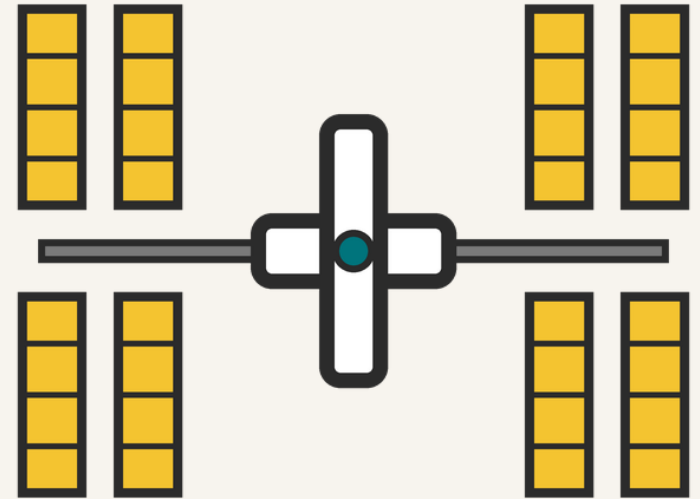
newton (N)

SPACE

Space Exploration

Getting there, staying alive, and
getting home again.

National 5 Physics (S3)



THE BIG QUESTION

People have lived up here without a break since November 2000. What out there would kill them in seconds?



The International Space Station — NASA

PART 1

Staying alive

Space is trying to kill you. Physics is how we stop it.

By the end of this part...

- 1 I can describe several challenges of space travel
- 2 I can describe the risks of crewed space exploration

Show me — Describe each word

SHOW ME BOARDS

1 the International Space Station

A crewed laboratory in orbit

2 radiation

Energy from the Sun and deep space

3 pressure

The push of the air on each unit of area

4 re-entry

Coming back into the atmosphere

5 vacuum

A space with no air in it

Show me — Describe each word

SHOW ME BOARDS

6 fuel

The store of energy a rocket burns

7 life support

The systems that keep a crew alive

8 solar cell

It transforms light into electrical energy

9 orbit

A path around a planet or star

10 radioactive decay

An unstable nucleus breaking down

Task 1 – life on the ISS

Booklet p.8

Watch the video about life on the ISS.
Write down every challenge you remember,
and the solution to each one.

Booklet p.8, Task 1

Task 1 – some answers

ANSWERS

Booklet p.8

Challenge	Solution
No air	oxygen made from water
Little water	water is recycled
Weak muscles and bones	exercise every day
Radiation	shielding in the walls
Very hot and very cold	insulation, heating, cooling

Weak muscles and bones

A challenge of living in orbit

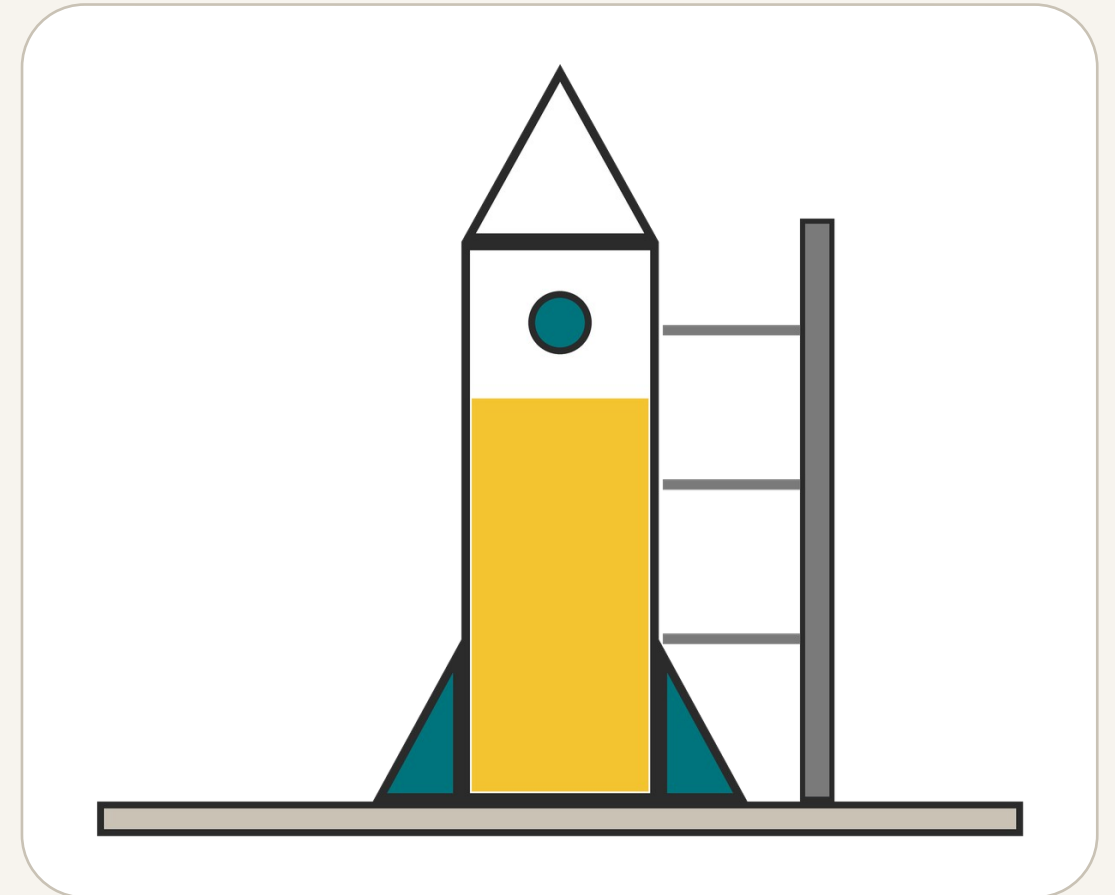


- In orbit nothing pushes back on your legs.
- Muscles and bones weaken without that load.
- Crews exercise for about two hours a day.
- Harnesses pull them down onto the treadmill.

Risk 1 – fuel load on take-off

Booklet p.8, Task 2

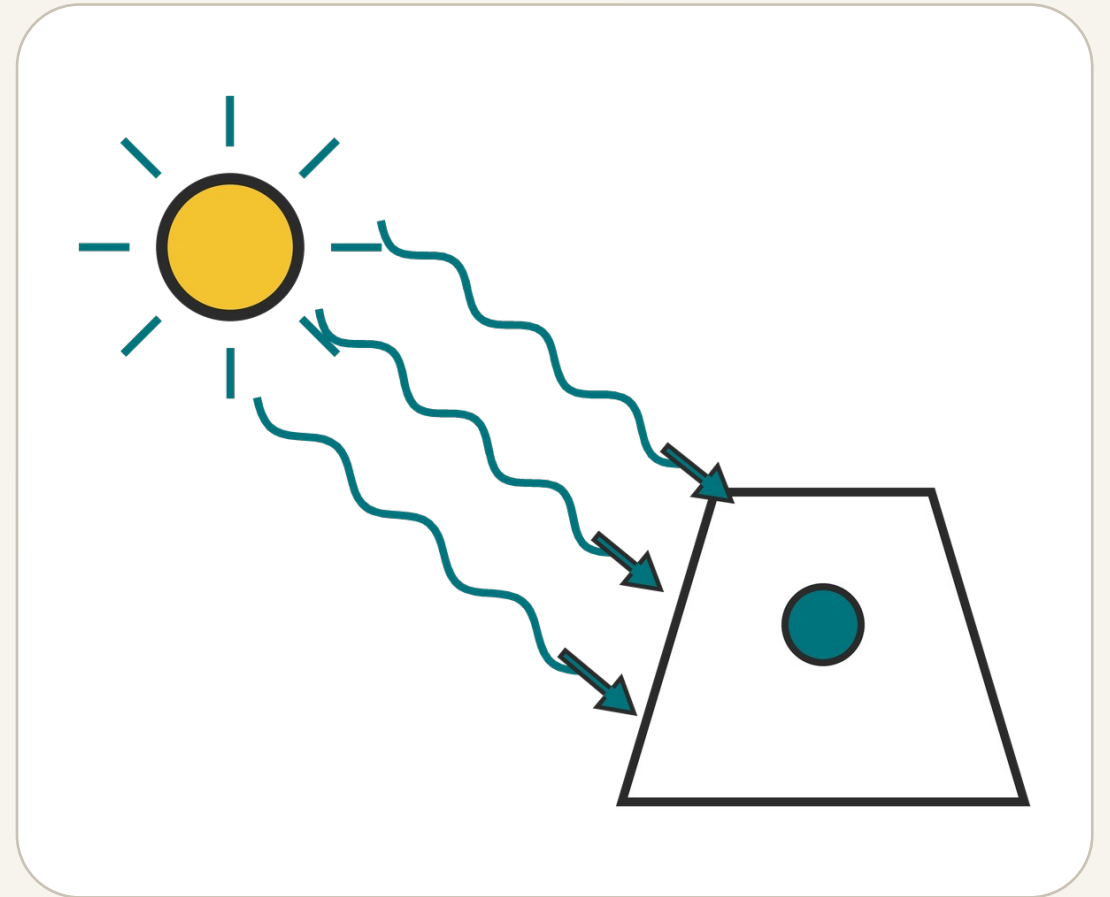
- Most of a rocket's mass at launch is fuel.
- More fuel means more weight to lift.
- Tonnes of fuel can explode.



Risk 2 – exposure to radiation

Booklet p.8, Task 2

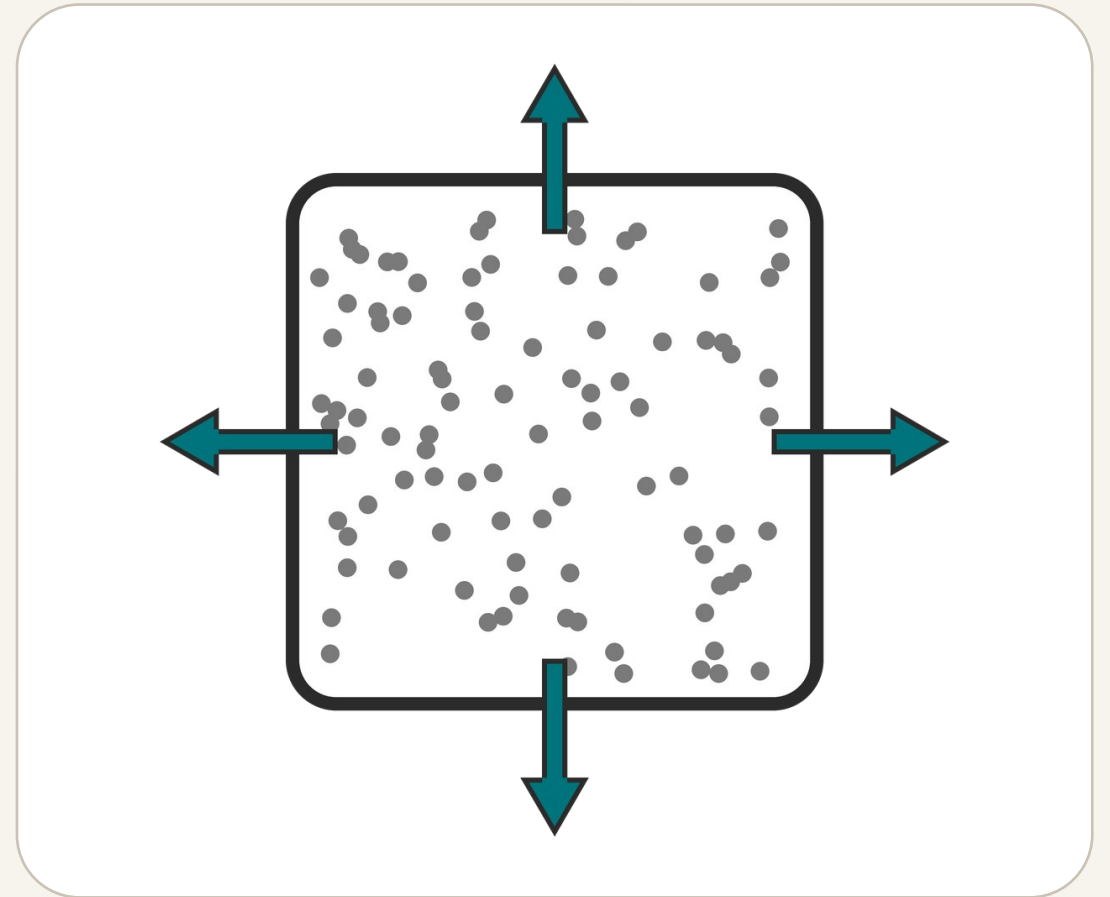
- Radiation reaches us from the Sun and from deep space.
- Above the atmosphere there is much less protection.
- Radiation damages the cells of the body.



Risk 3 – pressure differential

Booklet p.8, Task 2

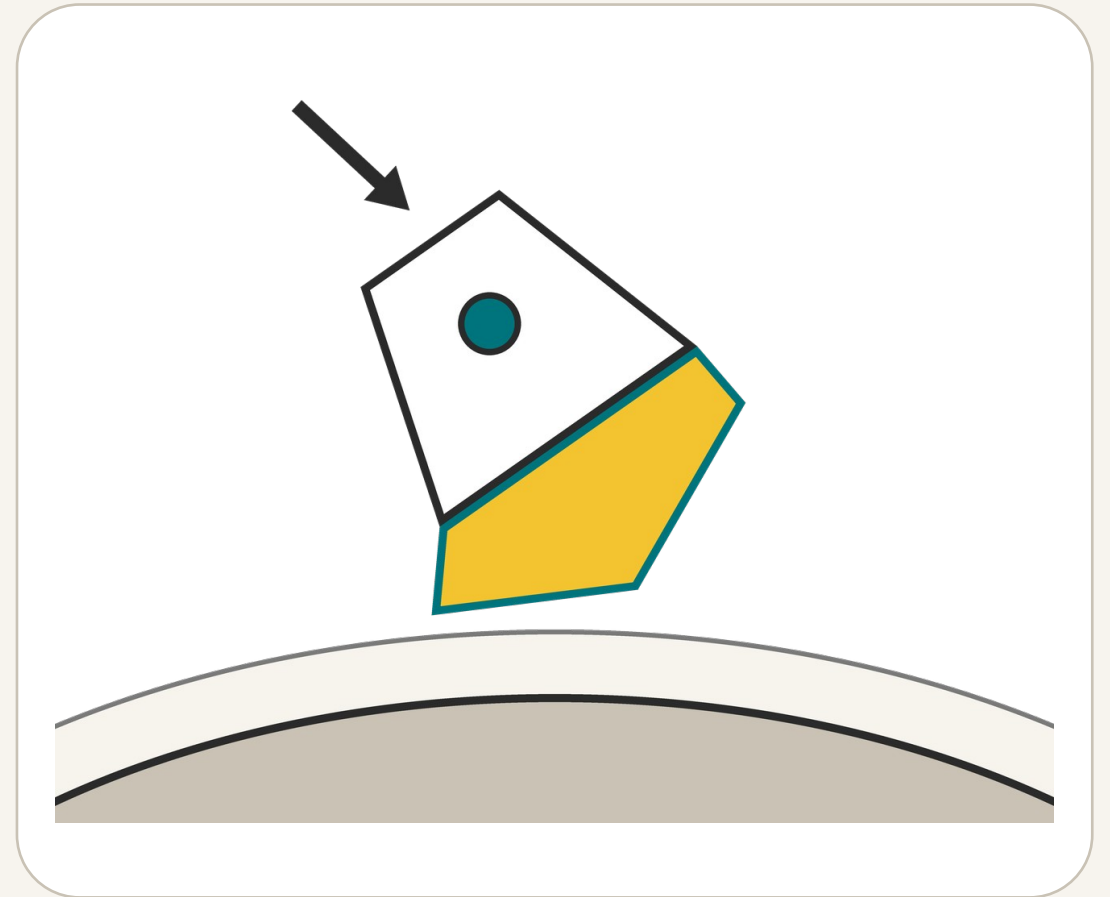
- Inside the craft: air at normal pressure.
- Outside: a vacuum, with no air at all.
- A leak lets the air rush out into space.



Risk 4 – re-entry

Booklet p.8, Task 2

- The craft enters the atmosphere very fast.
- Friction with the air heats it to very high temperatures.
- The heat shield must take the heat, at the right angle.



Artemis II comes home

Re-entry, for real



- April 2026: the first crew round the Moon since 1972.
- The capsule hits the atmosphere at enormous speed.
- Its heat shield takes the heat of re-entry.
- Parachutes then slow it for splashdown.

Show me — the risks

SHOW ME BOARDS

- 1 Why is a large fuel load a risk at take-off?
- 2 Where does the radiation astronauts meet come from?
- 3 What is on the other side of a spacecraft's wall?
- 4 What heats a craft as it re-enters?
- 5 Why must a spacesuit be pressurised?

Tonnes of fuel could
explode

The Sun and deep space

A vacuum — no air at all

Friction with the
atmosphere

To keep air around the
body

Show me — the risks

SHOW ME BOARDS

EXTRA PRACTICE

- 1 Which risk does a heat shield protect against?
- 2 Which risk is greatest in the first minutes?
- 3 Why is radiation a bigger risk above the air?
- 4 A small hole appears in the wall. What happens?
- 5 Name one way to shield from radiation.

Re-entry heating

The fuel load

There is less shielding

The air rushes out

Shielding in the walls

Correct your booklet

Booklet p.1 and p.8

CROSS OUT AND FIX

Success criteria: “give example of” should read “give examples of”.

Task 2, last risk: “earths” should read “Earth's”.

Task 2 – research the risks

Booklet p.8

Describe each risk in a couple of sentences.
Then say how the risk is reduced.

Booklet p.8, Task 2

Task 2 – Answers

ANSWERS

Booklet p.8 – risks 1 and 2

Risk	Description	How to reduce
Fuel load	tonnes of fuel that could explode	safety and crew escape systems
Radiation	damages body cells; less shelter above the air	shielding; shorter missions

Task 2 – Answers

ANSWERS

Booklet p.8 – risks 3 and 4

Risk	Description	How to reduce
Pressure	air inside, vacuum outside; a leak empties the cabin	strong sealed hull; pressurised suits
Re-entry	friction with the air makes the craft very hot	heat shield; the correct entry angle

Daily Review

Booklet p.8 – Part 1

- 1 Name two risks of crewed space exploration.
- 2 Where does space radiation come from?
- 3 Why is re-entry dangerous?
- 4 What is a pressure differential?
- 5 What protects a craft on re-entry?

Fuel, radiation,
pressure, re-entry

The Sun and deep space

Friction heats the craft

Air inside, vacuum
outside

The heat shield

PART 2

Keeping the power on

No mains sockets in orbit. Everything runs on what you brought.

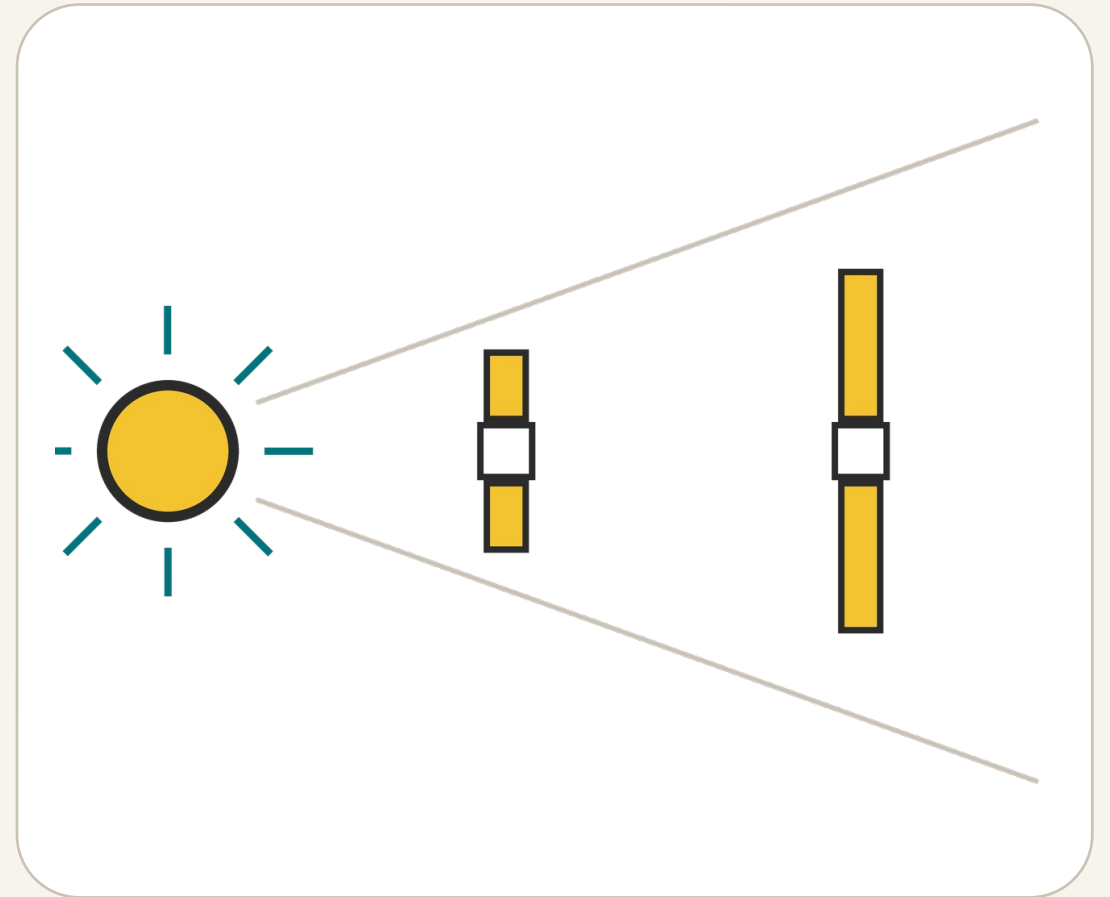
By the end of this part...

- 1 I can give methods of keeping life support systems powered
- 2 I can answer an open-ended question about space travel

Solar cells

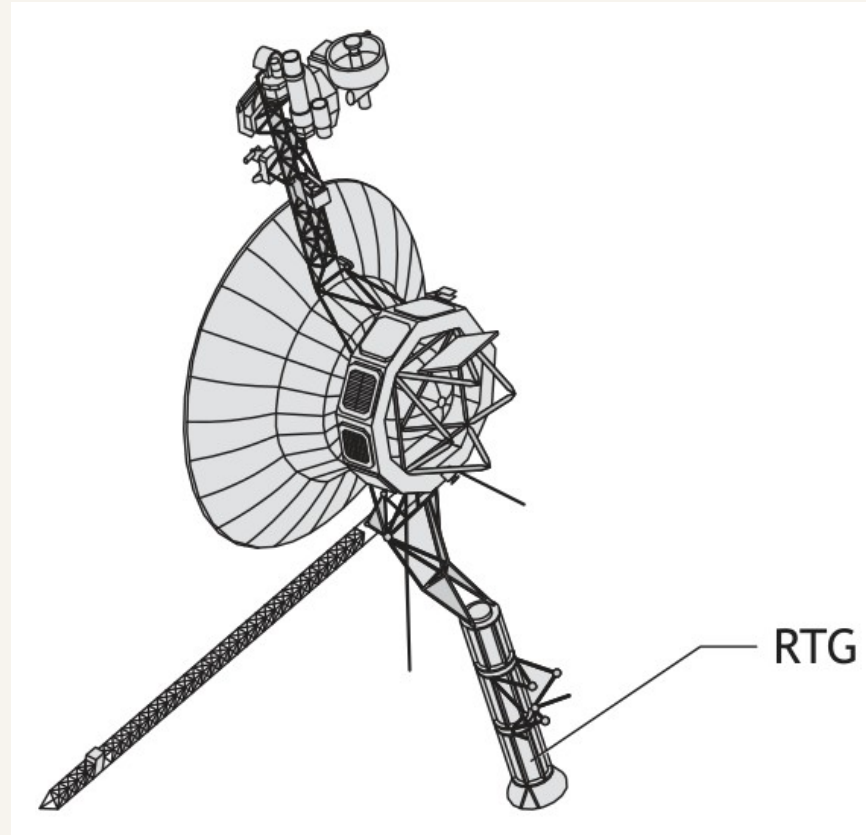
Booklet p.8, Task 3

- Solar cells transform light into electrical energy.
- Further from the Sun, the light is more spread out.
- So a craft further away needs a bigger area of cells.



A radioisotope thermoelectric generator

Booklet p.8, Task 3 — an RTG



Heat from radioactive decay is transformed into electrical energy.

Hinge question

FINGER VOTE

Think first — then show A to E on your fingers, 1 to 5

A probe is sent to Neptune, far from the Sun.
Its best supply of electrical energy is

- A** Solar cells — they never run out
- B** Solar cells — just make the area bigger
- C** An RTG — heat from radioactive decay
- D** An RTG — it collects the Sun's heat
- E** Batteries charged before launch.

Answer C

Show me — power in space

SHOW ME BOARDS

EXTRA PRACTICE

- 1 An RTG uses heat from what?
- 2 Why can solar cells not work in a planet's shadow?
- 3 Give one drawback of an RTG.
- 4 Probe B is further from the Sun. Which needs more cells?
- 5 Why does an RTG's power fall over the years?

Radioactive decay

No light reaches them

It is radioactive

Probe B

The fuel decays

Task 3 – keeping the power on

Booklet p.8

Describe solar cells and RTGs.
Give a benefit and a drawback of each.

Booklet p.8, Task 3

Task 3 – Answers

ANSWERS

Booklet p.8

	Benefits	Drawbacks
Solar cells	no fuel needed; nothing to run out	weak far from the Sun; useless in shadow
RTG	works far from the Sun and in shadow	radioactive; power falls as fuel decays

Would you rather...

TALK IT OUT

...spend a year on the ISS,
...or six months on the surface of the Moon?

Use the four risks as your reasons.
Which of them get worse on the Moon?

Task 4 – design a space station

Booklet p.9

In groups, design a station to live on for a year. Show three ways it beats the challenges of space, and explain the physics behind each one.

Booklet p.9, Task 4

Britons in space

Tim Peake, ESA — Expedition 47



- Helen Sharman went to Mir in 1991.
- Tim Peake: 186 days on the ISS, 2015-16.
- He ran the London Marathon up there.
- UK astronauts fly with ESA.

See it for yourself

TRY THIS TONIGHT

The ISS is bright enough to see with your eyes.
It crosses the sky in about four minutes.
Look for a steady bright “star” that does not blink.

NASA's Spot the Station gives the times where you live.

Command words – 1 of 2

Word	What it asks for	A Space example
State	Name it, or give it in brief form.	State two risks of crewed space travel.
Describe	Give its features: what happens.	Describe what happens to a craft on re-entry.
Explain	Link cause and effect.	Explain why a craft heats up on re-entry.

Command words – 2 of 2

Word	What it asks for	A Space example
Suggest	Use your physics in a new situation.	Suggest why probes to Neptune carry an RTG.
Predict	Say what may happen, from the facts given.	Predict the power from cells out at Jupiter.
Justify	Give reasons that back up your answer.	Solar cells or an RTG? Justify your answer.

What a 3-mark answer shows

Open-ended questions

- 3 marks: good understanding – the principles, a relationship or equation, applied. It need not be complete.
- 2 marks: reasonable understanding – relevant statements.
- 1 mark: limited understanding – a little of the physics.
- 0 marks: nothing relevant, or just repeating the question.

Open-ended practice

In your jotter (3 marks)

A pupil says that sending people to Mars is too dangerous to be worth it.

Using your knowledge of physics, comment on the risks the crew would face.



Checkpoint

- 1 Describe one risk of crewed space travel, and how it is reduced.
- 2 Why does a craft far from the Sun need a larger area of solar cells?
- 3 What is a pressure differential, and why is it dangerous?

e.g. re-entry heating —
a heat shield

The light is more spread
out

Air in, vacuum out: a
leak empties it

Exam practice

Past paper

- National 5 Physics 2024, Section 2, question 4(b). 1 mark.
- A physics challenge for astronauts on the surface of the Moon.
- 2024, Section 2, question 15(a). 1 mark. Power far from the Sun.
- Then booklet p.9, Task 5.

LOOKING AHEAD

Getting Around in Space

No air, no friction, no brakes. So how do you steer?

Plenary

- 1 The big question: what would kill you out there in seconds?
- 2 Name the four risks of crewed space exploration.
- 3 What does a heat shield protect a spacecraft from?
- 4 What energy transfer does a solar cell carry out?
- 5 Where does an RTG get its energy from?

The vacuum – no air and no pressure

Fuel load, radiation, pressure, re-entry

The heat of re-entry

Light to electrical energy

Radioactive decay

Plenary

EXTRA PRACTICE

- 1 Why is the fuel load a risk at take-off?
- 2 Why is radiation worse above the atmosphere?
- 3 What causes a spacecraft to heat up on re-entry?
- 4 Why do solar cells need a bigger area far out?
- 5 Give one benefit of an RTG.

Tonnes of fuel could explode

There is less shielding

Friction with the air

The light is more spread out

It works far from the Sun

Extension

- 1 Suggest why RTGs, not solar cells, power probes to Neptune.
- 2 A craft enters the atmosphere too steeply. Suggest what could happen.
- 3 Why is more fuel a problem as well as a help on a long mission?

Extension

EXTRA PRACTICE

- 1 Suggest why astronauts on a mission to Mars face more radiation.
- 2 Why must a spacesuit keep air pressure around the astronaut's body?
- 3 Suggest one danger of launching a spacecraft that carries an RTG.

Course specification – 1 of 3

N5 Physics course specification v4.0, p.8 – Space exploration

- Awareness of the challenges of space travel:
 - travelling large distances with the possible solution of attaining high velocity by using ion drive (producing a small unbalanced force over an extended period of time)
 - travelling large distances using a ‘catapult’ from a fast moving asteroid, moon or planet

Ion drive and the catapult are taught next lesson.

Course specification – 2 of 3

N5 Physics course specification v4.0, p.8 – Space exploration

- Awareness of the challenges of space travel:
 - manoeuvring a spacecraft in a zero friction environment, possibly to dock with the ISS
 - maintaining sufficient energy to operate life support systems in a spacecraft, with the possible solution of using solar cells with area that varies with distance from the Sun

Manoeuvring and docking are taught next lesson.

Course specification — 3 of 3

N5 Physics course specification v4.0, p.8 — Space exploration

- Awareness of the risks associated with manned space exploration:
 - fuel load on take-off
 - potential exposure to radiation
 - pressure differential
 - re-entry through an atmosphere